

FIE401: Exam

High-Frequency Trading and Stock Market Volatility

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Formalities

This final project will be handed out on **06.05.2020 at 9:00** and has to be submitted no later than the **08.05.2020 at 14:00** via WiseFlow.

In addition to the required hand-in files, there will be an oral exam on the **13.05.2020 – 15.05.2020**. Oral examination will be arranged via Zoom. Please check the presentation schedule on Canvas. You do not need to participate at the presentations of your peers. You will have strictly 5 minutes to present your solution which will be followed by a question and answer session lasting up to 20 minutes. The latter will refer to **both your analysis of the final project as well as any other topics covered in class**.

This exam is group-based. All group members will receive the same grade for the written submission, while grades on the oral exam will be individual.

Please submit the following documents:

- Coding file (.R): Please comment your general coding approach for each task. You do not need to explain the used functions.
- Report (.pdf): In the report, you should present, analyse, and interpret your results. You should provide numeric and written answers to all questions asked in the exam. Do not discuss your coding in the pdf as this should be found in the R file. Please keep

your answers short and precise and focus on the questions asked in the outline of the project.

- Presentation (.pdf): It is not allowed to change the slides between your submission and the actual presentation. Make sure that you use your 5 minutes presentation time wisely. It is important that you provide an economic rational for choices you made during this project.

Formal requirements

Report

- Motivate all your choices.
- Interpret your findings from a statistical and economical perspective.
- Your report should read like a mini thesis/article having following structure:
 1. Abstract - summarise your findings (wordlimit = 100 words)
 2. Intro - outline the research objective
 3. Data - summarize the data and variables used (Table 1)
 4. Analysis - your analysis (Table 2 & 3)
 5. Conclusion - your findings distilled
- The report should include three tables:
 1. Summary statistics
 2. Impact of the NYSE Hybrid Market introduction on high-frequency trading activity.
 3. Impact of high-frequency trading activity on the stock market volatility.
- The maximum word count of the report without title, tables, abstract, and header is 2,500.
- Make sure that tables are self-contained (use Guidelines for making proper tables available on Canvas).

- Present a maximum of six models in each of Tables 2 & 3.
- Use fairly plain formatting ensuring readability.
- No appendix.
- No figures.
- You do not need to explain the empirical models used as this will be part of the oral exam.

Presentation

- The main purpose of the presentation is to be a starting point for the oral exam.
- The time limit of 5 minutes will be strictly enforced.
- Discuss your choices of how you set up the econometric analysis.
- All three tables of the report have to be in the presentation.
- You do not need to prepare an appendix for the presentation.
- Do not write your names or any identifying information on the slides.

Your task

Modern equity markets are dominated by high-frequency traders (HFTs).¹ These traders are computer-based traders that are characterized by much faster reaction time and superior information processing technology than human beings. Stock exchanges around the globe implemented various technical solutions to be considered as fast markets and be attractive to HFTs.

For example, on March 22, 2006, the SEC approved the NYSE Hybrid Market proposal. The rollover of stocks to the NYSE Hybrid Market was implemented in a staggered way²

¹For popularized view see Lewis (2014).

²Stocks were transferred to the new system one by one. This implies that on each date some stocks were already trading on NYSE Hybrid market, while some stocks were still trading on the old system.

from October 2006 through January 2007. Though the NYSE had been testing new systems beforehand, the actual increase in automation and speed took place after October 2006.

NYSE Chief Technology Officer Steve Rubinow has commented on the tremendous execution speed increase achieved by the introduction of the NYSE Hybrid Market: *“The Hybrid Market launch has been very successful. With trade execution times [for Hybrid Market] decreasing from an average of nine seconds to less than 0.3 seconds, our customers and the trading floor community are using the system effectively throughout the day.”*³

High-frequency trading strategies are famous for submitting much more orders (quotes) relative to the actual number of executed trades. For example, SEC (2010) uses the quote-to-trade ratio as part of their HFT definition: *“Other characteristics often attributed to proprietary firms engaged in HFT are... (4) the submission of numerous orders that are cancelled shortly after submission...”* (p. 45).

Each high-frequency trader endogenously decides to which degree he/she wants to participate in a particular stock. This decision is based (though not limited to) the following stock characteristics: stock volatility, market capitalization, transaction costs, turnover, etc.

In this exam, you are investigating following research question:

Does high-frequency trading affect stock volatility?

Given the supplied data, you will have to settle for a set of dependent⁴ and independent variables. Make sure that your choices are motivated by economic intuition spelled out in your report.

Your analysis proceeds in three steps (each step is a table):

1. Summarize the data and variables used.
2. Investigate whether rollover to the NYSE Hybrid Market affects high-frequency trading activity.
3. Investigate a link between high-frequency trading activity and stock market volatility.

³See Eric Lai, “NYSE Launches New Hybrid Trading,” Computerworld, October 16, 2006.

⁴E.g., as a proxy for stock market volatility one can use historical standard deviation of stock returns, high minus low daily price range, etc.

Additional guidance

- Please use information and data as provided in the exam. In other words, do not gather additional data.
- Using code found on the internet is allowed, as long as you reference the sources correctly.
- Plagiarism in any way is not allowed.
- Do not be discouraged if data is fairly messy (i.e., requires additional cleaning), it usually is.
- Try to understand the data before you decide on an econometric approach.
- You need to make informed and motivated choices for variables and models used. Use economic intuition rather than maximizing statistical significance or R^2 .
- Make sure that your results are not driven by outliers in your suggested variables.
- Make sure that you decide on variables and data sets you want to use in your model, before starting to code.
- If you have questions during the exam, do not email me, but rather post your question under the respective thread on the Discussion Board on Canvas. This ensures that information is public and distributed fairly. Note that I will exercise my own judgement which questions to answer.
- We encourage students facing problems with R to utilize the Virtual Desktop Infrastructure that NHH offers.
- Good luck!

Data

1. HFT_proxy.csv is a data that includes information on two proxies for high-frequency trading activity. Following variables are included:
 - DATE: date (in yyyyymmdd format)
 - TICKER: firm identifier
 - QTE_DVOL: number of quote updates to the dollar volume executed per USD

10,000.

- QTE_TRD: number of quote updates to the number of trades.

2. Dates_Hybrid_intro.csv (data come from Terrence Hendershott's website)

- TICKER: firm identifier
- date: date (in yyyyymmdd format)
- EventTime: number of days relative to the rollover date to NYSE Hybrid market (EventTime=0 is a date when the stock was rolled over to the new system)

3. Data_from_CRSP.RData

- date: date (in yyyyymmdd format)
- TICKER: firm identifier
- SHRCD: a two-digit code describing the type of shares traded. Common shares are represented by codes 10 and 11
- EXCHCD: a code indicating the exchange on which a security is listed:
 - NYSE: 1 and 31
 - AMEX: 2 and 32
 - NASDAQ: 3 and 33
- BIDLO: is the lowest trading price during the day, or the closing bid price on days when the closing price is not available. The field is set to zero if no Bid or Low Price is available. If the PRC is negative, this field contains the closing bid. If positive, the field contains the lowest trade.
- ASKHI: is the highest trading price during the day, or the closing ask price on days when the closing price is not available. The field is set to zero if no Ask or High Price is available. If the PRC is negative, this field contains the closing ask. If positive, the field contains the highest trade.
- PRC: PRC is the closing price or the negative BID/ASK average for a trading day. If the closing price is not available on any given trading day, the number in

the price field has a negative sign to indicate that it is a bid/ask average and not an actual closing price. If neither closing price nor bid/ask average is available on a date, PRC is set to zero.

- VOL: number of shares traded.
- BID: closing bid price.
- ASK: closing ask price.
- SHROUT: number of publicly held shares, recorded in thousands.
- RET: return of the stock. Missing Return Codes: -66, -77, -88, -99.
- SICCD: Standard Industrial Classification code.

References

Lewis, M. (2014). Flash boys: A Wall Street Revolt.

SEC (2010). Concept release on equity market structure. *Release No. 34-61358, File No. S7-02-10.*